A thirteen-variable consistency equation

Strictly positive integer unknowns; exact degree 96; a 75-operation evaluator.

The only unknowns are

$$J, F, \alpha, f, i, T, s, w, n, \eta, \zeta, m, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
b = \beta J, \quad q = b + 1, \quad C = q - F - \alpha, \\
X = wq, \quad Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \\
\ell = EkY, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H, \\
R = [b(q - F) + \alpha](q^2 - 1) + (\mu + q\omega)J, \\
\tau = En + k - R, \quad L = cT - R, \quad V = f^2 L - c, \\
G = (K + w)C - F, \quad y = 1 + 4[b(m - LVG) - G], \\
P = \tau^2 - \ell(\ell + k), \quad Q = D^2 - \Delta c^2, \\
t = bic^2, \quad U = \Delta(f^2 - 1)(V^2 - y^2) + y^2, \\
Z = f^2 - \Delta t^2.
\end{gathered}$$

The single equation is

$$\boxed{(PQ)^2 + U^2 + Z^2 - 3 = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.
\end{gathered}$$

$$\begin{aligned}
\mu &= \beta - G_{951}(\rho) \\
&\quad - \rho^{960} G_{26}(\rho) - \rho^{1004}, \\
\omega &= \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda \\
&\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1004}, \\
K &= \rho^{18} + \rho^{1004} + \rho^{2020} \\
&\quad + 2\Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{gathered}
\chi_i = 1 + 2^{10}(1 + w_i + 2^8 n_i \\
+ 2^{17} p_i + 2^{25} s_i) \quad (0 \le i \le 941), \\
\pi(0) = 0, \quad \pi(i) = 1 + \sum\nolimits_{j=1}^{941} \left\lfloor \frac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).
\end{gathered}$$

$$\begin{aligned}
\Phi(x) = &\sum\nolimits_{i=0}^{941} \chi_i x^{\pi(i)} \\
&+ 2^{10} x^{969}(x^9 + 2^{17}) G_8(2x) \\
&+ 2^{18} x^{942}(x^{18} + 2^{17}) G_9(2x).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \lfloor \mathcal{A}/2^{636} \rfloor, \\
\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1 - \nu_j} \quad (0 \le j < 430), \\
\sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430), \\
m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14), \\
u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216, \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

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12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
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The corresponding Lean theorem concerns the selected formal encoding of Con(ZFC) and a proof in intuitionistic primitive recursive arithmetic. This page is a specification; see the project proof and its verification records for the exact theorem and encoding.